

Project Title

**Bankruptcy Prediction Using Decision Trees and Random
Forest**

**MACHINE LEARNING (25SC2107E)
in**

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Submitted by

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ABSTRACT

Bankruptcy Prediction is a Machine Learning-based classification project that aims to predict whether a company is likely to become bankrupt by analyzing its financial information. Early prediction of bankruptcy is essential for banks, investors, financial institutions, and business organizations, as it helps reduce financial losses and supports better decision-making. This project is based on Supervised Learning, where the model is trained using historical financial data containing features such as profit, debt, assets, liabilities, revenue, and cash flow. The project implements two popular classification algorithms, Decision Tree and Random Forest, to learn patterns from the dataset and classify companies as Bankrupt or Not Bankrupt. The performance of both models is evaluated and compared using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to determine the most reliable prediction model. Random Forest generally provides better accuracy by combining the predictions of multiple decision trees, reducing overfitting and improving overall performance. This project demonstrates the practical application of Machine Learning in financial risk analysis and highlights how predictive models can assist organizations in identifying financially vulnerable companies, enabling timely action and informed business decisions.

Signature of the Guide